

**LAPORAN PRAKTIKUM**  
**POSTTEST (3)**  
**ALGORITMA PEMROGRAMAN LANJUT**

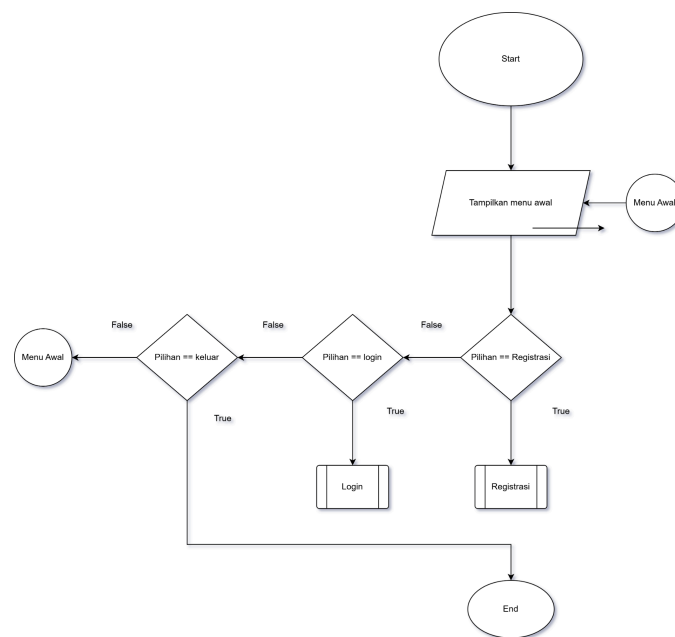


**Disusun oleh:**  
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**KELAS (C'1)**

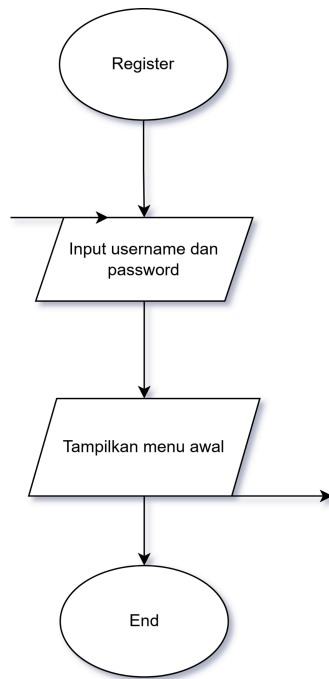
**PROGRAM STUDI INFORMATIKA**  
**UNIVERSITAS MULAWARMAN**  
**SAMARINDA**  
**2025**

## 1. Flowchart

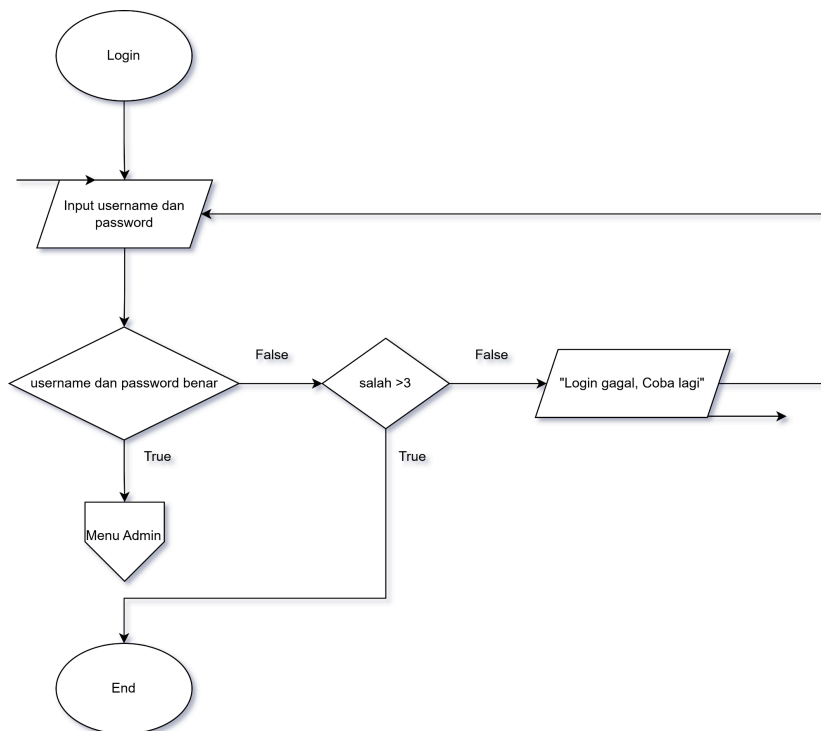
Flowchart ini membantu memvisualisasikan keputusan bercabang (pakai simbol belah ketupat) dan tindakan proses (pakai persegi panjang), sehingga mudah melihat jalur mana yang ditempuh sesuai kondisi.. Kemudian flowchart menggunakan predefined proses agar menunjukkan bahwa ada proses/fungsi yang terdefiniskan



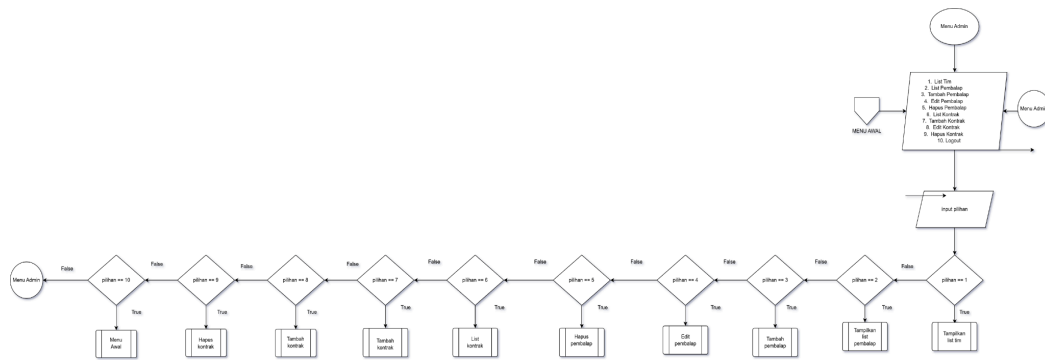
**Gambar 1.1 Flowchart CRUD Menu awal**



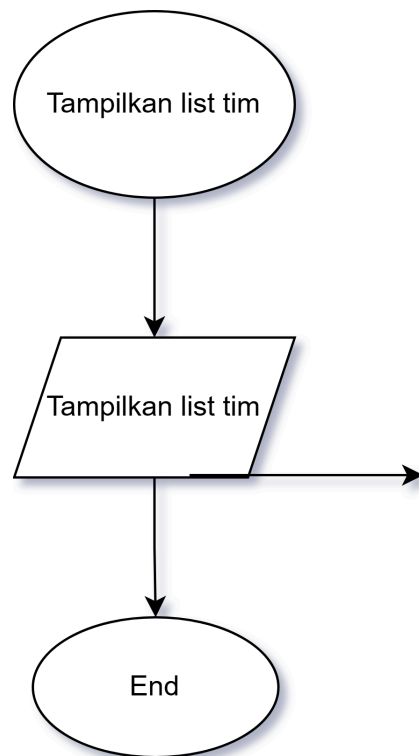
**Gambar 1.2 Flowchart Register**



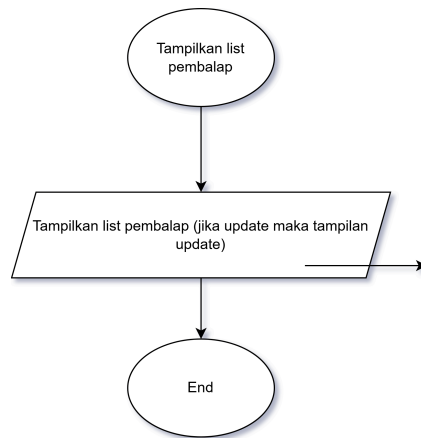
**Gambar 1.3 Flowchart Login**



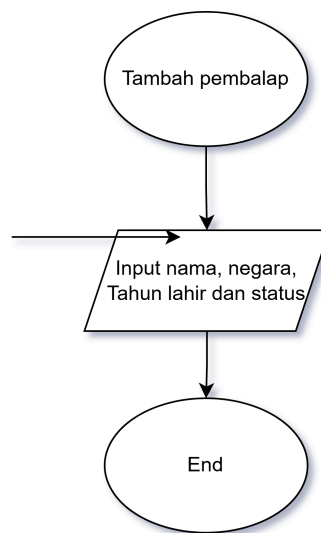
**Gambar 1.4 Flowchart Menu Admin**



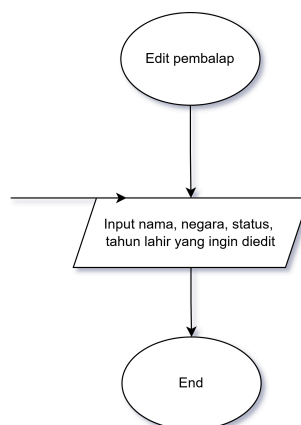
**Gambar 1.5 Tampilkan List Tim**



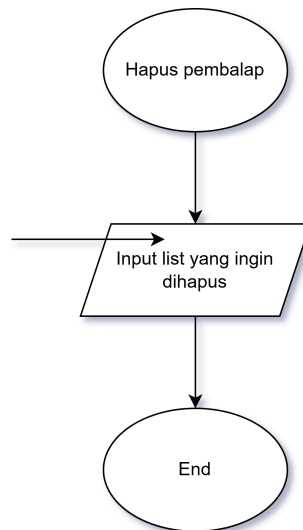
**Gambar 1.6 Tampilkan list pembalap**



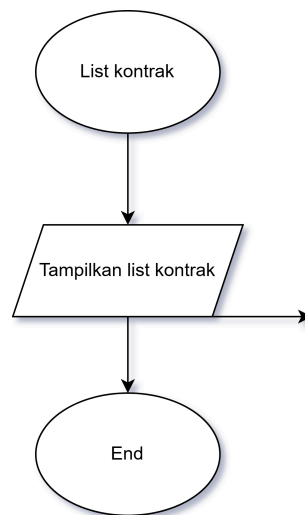
**Gambar 1.7 Tambah pembalap**



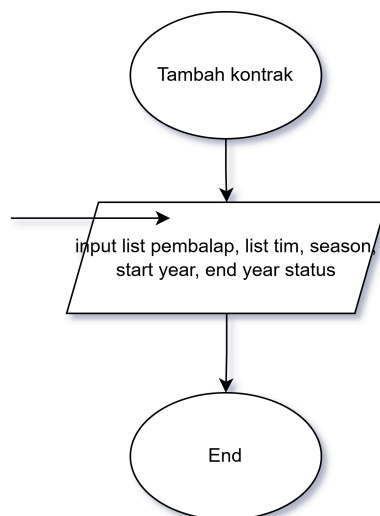
**Gambar 1.8 Edit Pembalap**



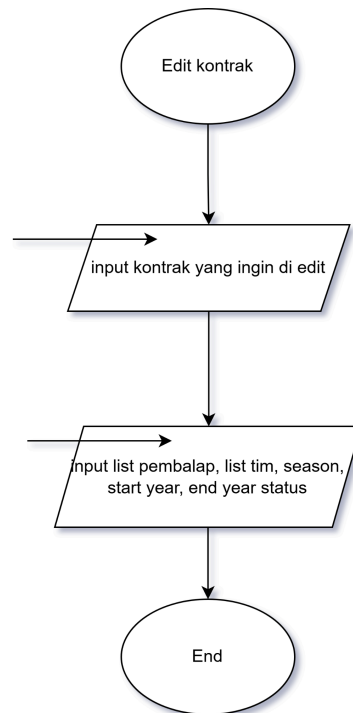
**Gambar 1.9 Hapus pembalap**



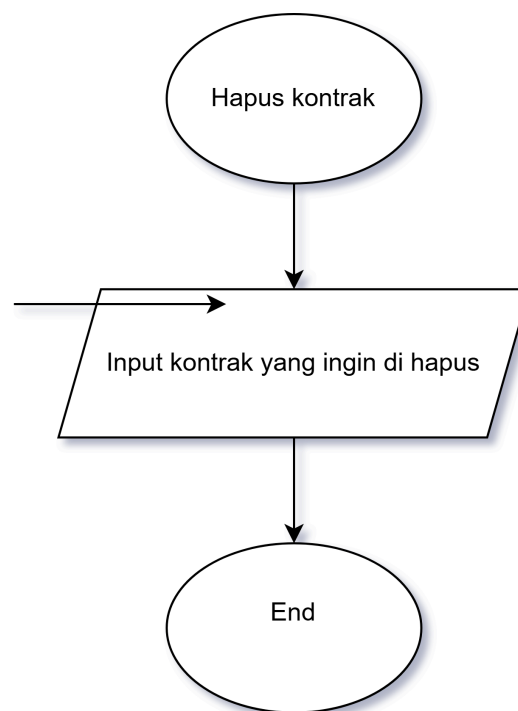
**Gambar 1.10 List Kontrak**



**Gambar 1.11 Tambah Kontrak**



**Gambar 1.12 Edit Kontrak**



**Gambar 1.13 Hapus kontrak**

## 2. Deskripsi Singkat Program

Secara keseluruhan, program ini berfungsi sebagai sistem manajemen kontrak dan informasi pembalap F1 yang sederhana, dengan kontrol akses berdasarkan role dan menu interaktif di konsol.

### 3. Source Code

```
#include <iostream>
#include <string>
#include <iomanip>
#include <cstdlib>
using namespace std;

const int MAX_USERS = 20;
const int MAX_DRIVERS = 50;
const int MAX_CONTRACTS = 100;
const int NUM_TEAMS = 11;
const int CURRENT_YEAR = 2026;

struct User {
    string username;
    string password;
};

struct Driver {
    string name;
    string nationality;
    int birthYear;
    string status;
};

struct Team {
    string name;
    bool active;
};

struct Contract {
    int driverIndex;
    int teamIndex;
    int season;
    string status;
    int startYear;
    int endYear;
};
```



```
void tampilJudul(string judul);
void tampilJudul(string judul, int garis);
void tampilGaris(int jumlah);

void menuAwal(User users[], int &userCount,
              Driver drivers[], int &driverCount,
              Team teams[], int teamCount,
              Contract contracts[], int &contractCount);

void menuAdmin(Driver drivers[], int &driverCount,
              Team teams[], int teamCount,
              Contract contracts[], int &contractCount);

void registerUser(User users[], int &userCount);
bool login(User users[], int userCount, int percobaan);

void listPembalap(Driver drivers[], int driverCount);
void listPembalapRekursif(Driver drivers[], int driverCount, int index);
void listTim(Team teams[], int teamCount);
void listKontrak(Contract contracts[], int contractCount,
                Driver drivers[], Team teams[]);

void tambahPembalap(Driver drivers[], int &driverCount);
void editPembalap(Driver drivers[], int driverCount);
void hapusPembalap(Driver drivers[], int &driverCount,
                  Contract contracts[], int &contractCount);

void tambahKontrak(Contract contracts[], int &contractCount,
                  Driver drivers[], int driverCount,
                  Team teams[], int teamCount);

void editKontrak(Contract contracts[], int contractCount,
                Driver drivers[], int driverCount,
                Team teams[], int teamCount);

void hapusKontrak(Contract contracts[], int &contractCount,
                 Driver drivers[], Team teams[]);

void tampilGaris(int jumlah) {
    for (int i = 0; i < jumlah; i++) {
```

```

        cout << "=";
    }
    cout << "\n";
}

void tampilJudul(string judul) {
    cout << "\n";
    tampilGaris(50);
    cout << judul << "\n";
    tampilGaris(50);
}

void tampilJudul(string judul, int garis) {
    cout << "\n";
    tampilGaris(garis);
    cout << judul << "\n";
    tampilGaris(garis);
}

void registerUser(User users[], int &userCount) {
    tampilJudul("REGISTER ADMIN");
    cout << "Username baru : ";
    getline(cin, users[userCount].username);
    cout << "Password baru : ";
    getline(cin, users[userCount].password);

    userCount++;
    cout << "Register berhasil.\n";
}

bool login(User users[], int userCount, int percobaan) {
    string username, password;

    cout << "Username : ";
    getline(cin, username);
    cout << "Password : ";
    getline(cin, password);

    for (int i = 0; i < userCount; i++) {
        if (users[i].username == username && users[i].password == password)
        {

```

```

        cout << "Login berhasil.\n";
        return true;
    }
}

percobaan++;

if (percobaan == 3) {
    cout << "Anda salah memasukkan username atau password sebanyak 3
kali. Program berhenti.\n";
    exit(0);
}

cout << "Username atau password salah. Coba lagi.\n";
return login(users, userCount, percobaan);
}

void listPembalapRekursif(Driver drivers[], int driverCount, int index) {
    if (index == driverCount) {
        return;
    }

    int umur = CURRENT_YEAR - drivers[index].birthYear;

    cout << "| "
        << setw(3) << left << index + 1
        << "| " << setw(25) << left << drivers[index].name
        << "| " << setw(15) << left << drivers[index].nationality
        << "| " << setw(11) << left << drivers[index].birthYear
        << "| " << setw(5) << left << umur
        << "| " << setw(12) << left << drivers[index].status
        << "\n";

    listPembalapRekursif(drivers, driverCount, index + 1);
}

void listPembalap(Driver drivers[], int driverCount) {
    if (driverCount == 0) {
        cout << "Belum ada pembalap yang di tambahkan\n";
        return;
    }
}

```





```

cin.ignore();

cout << "Pilih status:\n";
cout << "1. Aktif\n";
cout << "2. Tidak Aktif\n";
cout << "Pilihan      : ";
cin >> pilihanStatus;
cin.ignore();

if (pilihanStatus == 1) {
    drivers[driverCount].status = "Aktif";
} else {
    drivers[driverCount].status = "Tidak Aktif";
}

driverCount++;
cout << "Pembalap berhasil ditambahkan.\n";
}

void editPembalap(Driver drivers[], int driverCount) {
    if (driverCount == 0) {
        cout << "Belum ada pembalap yang di tambahkan\n";
        return;
    }

    int pilih, pilihanStatus;

    listPembalap(drivers, driverCount);
    cout << "Pilih nomor pembalap yang ingin diedit : ";
    cin >> pilih;
    cin.ignore();

    pilih--;

    cout << "Nama baru      : ";
    getline(cin, drivers[pilih].name);
    cout << "Negara baru      : ";
    getline(cin, drivers[pilih].nationality);
    cout << "Tahun lahir baru : ";
    cin >> drivers[pilih].birthYear;
    cin.ignore();

```

```

        cout << "Pilih status baru:\n";
        cout << "1. Aktif\n";
        cout << "2. Tidak Aktif\n";
        cout << "Pilihan          : ";
        cin >> pilihanStatus;
        cin.ignore();

        if (pilihanStatus == 1) {
            drivers[pilih].status = "Aktif";
        } else {
            drivers[pilih].status = "Tidak Aktif";
        }

        cout << "Pembalap berhasil diedit.\n";
    }

void hapusPembalap(Driver drivers[], int &driverCount,
                   Contract contracts[], int &contractCount) {
    if (driverCount == 0) {
        cout << "Belum ada pembalap yang di tambahkan\n";
        return;
    }

    int pilih;

    listPembalap(drivers, driverCount);
    cout << "Pilih nomor pembalap yang ingin dihapus : ";
    cin >> pilih;
    cin.ignore();

    pilih--;

    for (int i = 0; i < contractCount; ) {
        if (contracts[i].driverIndex == pilih) {
            for (int j = i; j < contractCount - 1; j++) {
                contracts[j] = contracts[j + 1];
            }
            contractCount--;
        } else {
            if (contracts[i].driverIndex > pilih) {

```

```

        contracts[i].driverIndex--;
    }
    i++;
}
}

for (int i = pilih; i < driverCount - 1; i++) {
    drivers[i] = drivers[i + 1];
}

driverCount--;
cout << "Pembalap berhasil dihapus.\n";
}

void tambahKontrak(Contract contracts[], int &contractCount,
                  Driver drivers[], int driverCount,
                  Team teams[], int teamCount) {
    int d, t, pilihanStatus;

    tampilJudul("TAMBAH KONTRAK");

    listPembalap(drivers, driverCount);
    cout << "Pilih nomor driver : ";
    cin >> d;
    cin.ignore();

    listTim(teams, teamCount);
    cout << "Pilih nomor team : ";
    cin >> t;
    cin.ignore();

    contracts[contractCount].driverIndex = d - 1;
    contracts[contractCount].teamIndex = t - 1;

    cout << "Season : ";
    cin >> contracts[contractCount].season;
    cin.ignore();

    cout << "Pilih status kontrak:\n";
    cout << "1. Rumor\n";
    cout << "2. Resmi\n";
}

```



```

        cout << "Pilihan          : ";
        cin >> pilihanStatus;
        cin.ignore();

        if (pilihanStatus == 1) {
            contracts[contractCount].status = "Rumor";
        } else {
            contracts[contractCount].status = "Resmi";
        }

        cout << "Start Year          : ";
        cin >> contracts[contractCount].startYear;
        cin.ignore();
        cout << "End Year            : ";
        cin >> contracts[contractCount].endYear;
        cin.ignore();

        contractCount++;
        cout << "Kontrak berhasil ditambahkan.\n";
    }

    void editKontrak(Contract contracts[], int contractCount,
                    Driver drivers[], int driverCount,
                    Team teams[], int teamCount) {
        if (contractCount == 0) {
            cout << "Belum ada kontrak yang di tambahkan\n";
            return;
        }

        int pilih, d, t, pilihanStatus;

        listKontrak(contracts, contractCount, drivers, teams);
        cout << "Pilih nomor kontrak yang ingin diedit : ";
        cin >> pilih;
        cin.ignore();

        pilih--;

        listPembalap(drivers, driverCount);
        cout << "Driver baru : ";
        cin >> d;

```

```

    cin.ignore();

    listTim(teams, teamCount);
    cout << "Team baru      : ";
    cin >> t;
    cin.ignore();

    contracts[pilih].driverIndex = d - 1;
    contracts[pilih].teamIndex = t - 1;

    cout << "Season baru          : ";
    cin >> contracts[pilih].season;
    cin.ignore();

    cout << "Pilih status kontrak baru:\n";
    cout << "1. Rumor\n";
    cout << "2. Resmi\n";
    cout << "Pilihan                : ";
    cin >> pilihanStatus;
    cin.ignore();

    if (pilihanStatus == 1) {
        contracts[pilih].status = "Rumor";
    } else {
        contracts[pilih].status = "Resmi";
    }

    cout << "Start Year baru      : ";
    cin >> contracts[pilih].startYear;
    cin.ignore();
    cout << "End Year baru        : ";
    cin >> contracts[pilih].endYear;
    cin.ignore();

    cout << "Kontrak berhasil diedit.\n";
}

void hapusKontrak(Contract contracts[], int &contractCount,
                  Driver drivers[], Team teams[]) {
    if (contractCount == 0) {
        cout << "Belum ada kontrak yang di tambahkan\n";
    }
}

```

```

        return;
    }

    int pilih;

    listKontrak(contracts, contractCount, drivers, teams);
    cout << "Pilih nomor kontrak yang ingin dihapus : ";
    cin >> pilih;
    cin.ignore();

    pilih--;

    for (int i = pilih; i < contractCount - 1; i++) {
        contracts[i] = contracts[i + 1];
    }

    contractCount--;
    cout << "Kontrak berhasil dihapus.\n";
}

void menuAdmin(Driver drivers[], int &driverCount,
               Team teams[], int teamCount,
               Contract contracts[], int &contractCount) {
    int pilih;

    do {
        tampilJudul("MENU ADMIN");
        cout << "+-----+\n";
        cout << "| 1. List Tim | \n";
        cout << "| 2. List Pembalap | \n";
        cout << "| 3. Tambah Pembalap | \n";
        cout << "| 4. Edit Pembalap | \n";
        cout << "| 5. Hapus Pembalap | \n";
        cout << "| 6. List Kontrak | \n";
        cout << "| 7. Tambah Kontrak | \n";
        cout << "| 8. Edit Kontrak | \n";
        cout << "| 9. Hapus Kontrak | \n";
        cout << "| 10. Logout | \n";
        cout << "+-----+\n";
        cout << "Pilihan : ";
        cin >> pilih;
    }
}

```

```

        cin.ignore();

        switch (pilih) {
            case 1: listTim(teams, teamCount); break;
            case 2: listPembalap(drivers, driverCount); break;
            case 3: tambahPembalap(drivers, driverCount); break;
            case 4: editPembalap(drivers, driverCount); break;
            case 5: hapusPembalap(drivers, driverCount, contracts,
contractCount); break;
            case 6: listKontrak(contracts, contractCount, drivers, teams);
break;
            case 7: tambahKontrak(contracts, contractCount, drivers,
driverCount, teams, teamCount); break;
            case 8: editKontrak(contracts, contractCount, drivers,
driverCount, teams, teamCount); break;
            case 9: hapusKontrak(contracts, contractCount, drivers, teams);
break;
            case 10: cout << "Logout.\n"; break;
            default: cout << "Pilihan tidak valid.\n";
        }
    } while (pilih != 10);
}

void menuAwal(User users[], int &userCount,
    Driver drivers[], int &driverCount,
    Team teams[], int teamCount,
    Contract contracts[], int &contractCount) {
    int pilih;

    do {
        cout << "+-----+\n";
        cout << "|          SELAMAT DATANG DI BURSA TRANSFER F1    |\n";
        cout << "+-----+\n";

        tampilJudul("MENU AWAL");
        cout << "+-----+\n";
        cout << "| 1.  Register Admin                                |\n";
        cout << "| 2.  Login Admin                                   |\n";
        cout << "| 3.  Keluar                                        |\n";
        cout << "+-----+\n";
        cout << "Pilihan : ";
    }
}

```

```

        cin >> pilih;
        cin.ignore();

        switch (pilih) {
            case 1:
                registerUser(users, userCount);
                break;
            case 2:
                if (login(users, userCount, 0)) {
                    menuAdmin(drivers, driverCount, teams, teamCount,
contracts, contractCount);
                }
                break;
            case 3:
                cout << "Program selesai.\n";
                break;
            default:
                cout << "Pilihan tidak valid.\n";
        }
    } while (pilih != 3);
}

int main() {
    User users[MAX_USERS] = {
        {"Attol", "2509106103"}
    };
    int userCount = 1;

    Driver drivers[MAX_DRIVERS] = {
        {"Charles Leclerc", "Monaco", 1997, "Aktif"},
        {"Lewis Hamilton", "UK", 1985, "Aktif"},
        {"George Russell", "UK", 1998, "Aktif"},
        {"Kimi Antonelli", "Italy", 2006, "Aktif"},
        {"Lando Norris", "UK", 1999, "Aktif"},
        {"Oscar Piastri", "Australia", 2001, "Aktif"},
        {"Fernando Alonso", "Spain", 1981, "Aktif"},
        {"Lance Stroll", "Canada", 1998, "Aktif"},
        {"Esteban Ocon", "France", 1996, "Aktif"},
        {"Oliver Bearman", "UK", 2005, "Aktif"},
        {"Isack Hadjar", "France", 2003, "Aktif"},
        {"Max Verstappen", "Netherlands", 1997, "Aktif"},
    };
}

```

```

        {"Carlos Sainz Jr.", "Spain", 1994, "Aktif"},
        {"Alexander Albon", "UK", 1996, "Aktif"},
        {"Valtteri Bottas", "Finland", 1989, "Aktif"},
        {"Sergio Perez", "Mexico", 1990, "Aktif"},
        {"Nico Hulkenberg", "Germany", 1987, "Aktif"},
        {"Gabriel Bortoleto", "Brazil", 2004, "Aktif"},
        {"Pierre Gasly", "France", 1996, "Aktif"},
        {"Franco Colapinto", "Argentina", 1999, "Aktif"},
        {"Liam Lawson", "New Zealand", 2002, "Aktif"},
        {"Arvid Lindblad", "Sweden", 2004, "Aktif"}
    };

    int driverCount = 22;

    Team teams[NUM_TEAMS] = {
        {"Scuderia Ferrari HP", true},
        {"Mercedes AMG-Petronas", true},
        {"Oracle RedBull Racing", true},
        {"McLaren Mastercard", true},
        {"Aston Martin Aramco", true},
        {"BWT Alpine", true},
        {"Atlassian Williams", true},
        {"TGR Haas", true},
        {"VCAB Racing Bull", true},
        {"Audi Revolut", true},
        {"Cadillac", true}
    };

    Contract contracts[MAX_CONTRACTS];
    int contractCount = 0;

    menuAwal(users, userCount, drivers, driverCount, teams, NUM_TEAMS,
contracts, contractCount);

    return 0;
}

```

#### 4. Hasil Output

```
+-----+
|      SELAMAT DATANG DI BURSA TRANSFER F1      |
+-----+

=====
MENU AWAL
=====
+-----+
| 1. Register Admin |
| 2. Login Admin   |
| 3. Keluar        |
+-----+
Pilihan : 2
Username : Attol
Password : 2509106103
Login berhasil.
```

Gambar 4.1 Hasil Output Menu Awal

```
=====
MENU ADMIN
=====
+-----+
| 1. List Tim       |
| 2. List Pembalap |
| 3. Tambah Pembalap |
| 4. Edit Pembalap |
| 5. Hapus Pembalap |
| 6. List Kontrak  |
| 7. Tambah Kontrak |
| 8. Edit Kontrak  |
| 9. Hapus Kontrak |
| 10. Logout       |
+-----+
Pilihan : █
```

Gambar 4.2 Hasil Output Login Sebagai Admin

```

+-----+
Pilihan : 1

=====
DAFTAR TIM
=====
+-----+
| No | Nama Tim                | Status |
+-----+
| 1  | Scuderia Ferrari HP    | Active |
| 2  | Mercedes AMG-Petronas  | Active |
| 3  | Oracle RedBull Racing  | Active |
| 4  | McLaren Mastercard     | Active |
| 5  | Aston Martin Aramco   | Active |
| 6  | BWT Alpine             | Active |
| 7  | Atlassian Williams     | Active |
| 8  | TGR Haas               | Active |
| 9  | VCAB Racing Bull       | Active |
| 10 | Audi Revolut            | Active |
| 11 | Cadillac               | Active |
+-----+

```

**Gambar 4.3 Hasil Output List Tim**

```

=====
DATA PEMBALAP LENGKAP
=====
+-----+
| No | Nama                    | Negara      | Tahun Lahir | Umur | Status |
+-----+
| 1  | Charles Leclerc        | Monaco      | 1997        | 29   | Aktif  |
| 2  | Lewis Hamilton         | UK          | 1985        | 41   | Aktif  |
| 3  | George Russell         | UK          | 1998        | 28   | Aktif  |
| 4  | Kimi Antonelli         | Italy       | 2006        | 20   | Aktif  |
| 5  | Lando Norris           | UK          | 1999        | 27   | Aktif  |
| 6  | Oscar Piastri          | Australia   | 2001        | 25   | Aktif  |
| 7  | Fernando Alonso        | Spain       | 1981        | 45   | Aktif  |
| 8  | Lance Stroll           | Canada     | 1998        | 28   | Aktif  |
| 9  | Esteban Ocon           | France     | 1996        | 30   | Aktif  |
| 10 | Oliver Bearman         | UK          | 2005        | 21   | Aktif  |
| 11 | Isack Hadjar           | France     | 2003        | 23   | Aktif  |
| 12 | Max Verstappen         | Netherlands | 1997        | 29   | Aktif  |
| 13 | Carlos Sainz Jr.       | Spain      | 1994        | 32   | Aktif  |
| 14 | Alexander Albon        | UK          | 1996        | 30   | Aktif  |
| 15 | Valtteri Bottas        | Finland    | 1989        | 37   | Aktif  |
| 16 | Sergio Perez           | Mexico     | 1990        | 36   | Aktif  |
| 17 | Nico Hulkenberg        | Germany    | 1987        | 39   | Aktif  |
| 18 | Gabriel Bortoleto      | Brazil     | 2004        | 22   | Aktif  |
| 19 | Pierre Gasly           | France     | 1996        | 30   | Aktif  |
| 20 | Franco Colapinto       | Argentina  | 1999        | 27   | Aktif  |
| 21 | Liam Lawson            | New Zealand| 2002        | 24   | Aktif  |
| 22 | Arvid Lindblad         | Sweden     | 2004        | 22   | Aktif  |
+-----+

```

**Gambar 4.4 Hasil Output List pembalap**



```

=====
TAMBAH PEMBALAP
=====
Nama      : Benjamin Seskowi
Negara    : Solovenia
Tahun lahir : 2002
Pilih status:
1. Aktif
2. Tidak Aktif
Pilihan    : 2
Pembalap berhasil ditambahkan.

```

**Gambar 4.5 Output Create Pembalap**

```

| 21 | Liam Lawson      | New Zealand | 2002 | 24 | Aktif |
| 22 | Arvid Lindblad   | Sweden      | 2004 | 22 | Aktif |
| 23 | Benjamin Seskowi | Solovenia   | 2002 | 24 | Tidak Aktif |
+---+-----+-----+-----+---+
Pilih nomor driver : 23

=====
DAFTAR TIM
=====
+---+-----+-----+
| No | Nama Tim          | Status |
+---+-----+-----+
| 1  | Scuderia Ferrari HP | Active |
| 2  | Mercedes AMG-Petronas | Active |
| 3  | Oracle RedBull Racing | Active |
| 4  | McLaren Mastercard   | Active |
| 5  | Aston Martin Aramco  | Active |
| 6  | BWT Alpine           | Active |
| 7  | Atlassian Williams    | Active |
| 8  | TGR Haas              | Active |
| 9  | VCAB Racing Bull     | Active |
| 10 | Audi Revolut         | Active |
| 11 | Cadillac              | Active |
+---+-----+-----+
Pilih nomor team   : 10
Season             : 2039
Pilih status kontrak:
1. Rumor
2. Resmi
Pilihan            : 1
Start Year         : 2040
End Year           : 2047
Kontrak berhasil ditambahkan.

```

**Gambar 4.6 Create Kontrak**

```
=====
DATA PEMBALAP LENGKAP
=====
```

| No | Nama              | Negara      | Tahun Lahir | Umur | Status      |
|----|-------------------|-------------|-------------|------|-------------|
| 1  | Charles Leclerc   | Monaco      | 1997        | 29   | Aktif       |
| 2  | Lewis Hamilton    | UK          | 1985        | 41   | Aktif       |
| 3  | George Russell    | UK          | 1998        | 28   | Aktif       |
| 4  | Kimi Antonelli    | Italy       | 2006        | 20   | Aktif       |
| 5  | Lando Norris      | UK          | 1999        | 27   | Aktif       |
| 6  | Oscar Piastri     | Australia   | 2001        | 25   | Aktif       |
| 7  | Fernando Alonso   | Spain       | 1981        | 45   | Aktif       |
| 8  | Lance Stroll      | Canada      | 1998        | 28   | Aktif       |
| 9  | Esteban Ocon      | France      | 1996        | 30   | Aktif       |
| 10 | Oliver Bearman    | UK          | 2005        | 21   | Aktif       |
| 11 | Isack Hadjar      | France      | 2003        | 23   | Aktif       |
| 12 | Max Verstappen    | Netherlands | 1997        | 29   | Aktif       |
| 13 | Carlos Sainz Jr.  | Spain       | 1994        | 32   | Aktif       |
| 14 | Alexander Albon   | UK          | 1996        | 30   | Aktif       |
| 15 | Valtteri Bottas   | Finland     | 1989        | 37   | Aktif       |
| 16 | Sergio Perez      | Mexico      | 1990        | 36   | Aktif       |
| 17 | Nico Hulkenberg   | Germany     | 1987        | 39   | Aktif       |
| 18 | Gabriel Bortoleto | Brazil      | 2004        | 22   | Aktif       |
| 19 | Pierre Gasly      | France      | 1996        | 30   | Aktif       |
| 20 | Franco Colapinto  | Argentina   | 1999        | 27   | Aktif       |
| 21 | Liam Lawson       | New Zealand | 2002        | 24   | Aktif       |
| 22 | Arvid Lindblad    | Sweden      | 2004        | 22   | Aktif       |
| 23 | Benjamin Seskowi  | Solovenia   | 2002        | 24   | Tidak Aktif |

```
=====
Pilih nomor pembalap yang ingin dihapus : 23
Pembalap berhasil dihapus.
```

#### 4.7 Output Delete Pembalap

```
=====
DAFTAR KONTRAK
=====
```

| No | Driver         | Team                | Season | Status | Start Year | End Year |
|----|----------------|---------------------|--------|--------|------------|----------|
| 1  | Arvid Lindblad | Scuderia Ferrari HP | 2029   | Rumor  | 2030       | 2031     |

```
=====
Pilih nomor kontrak yang ingin dihapus : 1
Kontrak berhasil dihapus.
```

Gambar 4.8 Hasil Output Delete kontrak

|    |                |        |      |    |       |
|----|----------------|--------|------|----|-------|
| 22 | Arvid Lindblad | Sweden | 2004 | 22 | Aktif |
|----|----------------|--------|------|----|-------|

---

Pilih nomor pembalap yang ingin diedit : 22  
 Nama baru : Arvid Limbad  
 Negara baru : Swedia  
 Tahun lahir baru : 2007  
 Pilih status baru:  
 1. Aktif  
 2. Tidak Aktif  
 Pilihan : 1  
 Pembalap berhasil diedit.

**Gambar 4.9 Edit Pembalap**

Driver baru : 1

---

DAFTAR TIM

---

| No | Nama Tim              | Status |
|----|-----------------------|--------|
| 1  | Scuderia Ferrari HP   | Active |
| 2  | Mercedes AMG-Petronas | Active |
| 3  | Oracle RedBull Racing | Active |
| 4  | McLaren Mastercard    | Active |
| 5  | Aston Martin Aramco   | Active |
| 6  | BWT Alpine            | Active |
| 7  | Atlassian Williams    | Active |
| 8  | TGR Haas              | Active |
| 9  | VCAB Racing Bull      | Active |
| 10 | Audi Revolut          | Active |
| 11 | Cadillac              | Active |

---

Team baru : 3  
 Season baru : 2030  
 Pilih status kontrak baru:  
 1. Rumor  
 2. Resmi  
 Pilihan : 2  
 Start Year baru : 2031  
 End Year baru : 2035  
 Kontrak berhasil diedit.

#### 4.10 Edit kontrak

## 5. Langkah-langkah GIT

```
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git add .
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git commit -m "KOMAT KAMIT"
[main aede32e] KOMAT KAMIT
3 files changed, 233 insertions(+)
create mode 100644 post-test/post-test-7/2509106103-KhayzanDwittraAttala-PT-7 (1).pdf
create mode 100644 post-test/post-test-7/2509106103-PT-7.drawio.png
create mode 100644 post-test/post-test-7/2509106103_KhayzanDwittraAttala-PT-7.py
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 868.02 KiB | 27.13 MiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Khayzannn/praktikum-apd.git
609b85f..aede32e main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> █
```

**Gambar 5.1 Langkah - Langkah Git**

### 5.1 GIT Add

Git add adalah perintah diGit yang digunakan untuk menambahkan perubahan pada berkas di direktori kerja ke area staging (atau indeks), mempersiapkannya untuk dimasukkan ke dalam commit berikutnya.

### 5.2 GIT Commit

Git commit adalah perintah di sistem kontrol versi Git untuk menyimpan (melakukan "commit") serangkaian perubahan yang telah dipilih (di-staging) ke dalam repositori lokal Anda, membuat sebuah "snapshot" atau titik pemeriksaan pada riwayat proyek dengan deskripsi perubahan.

### 5.3 GIT Push

Git push adalah perintah dalam Git yang digunakan untuk mengunggah (mengirimkan) perubahan yang telah dicatat di repositori lokal Anda ke repositori jarak jauh (seperti GitHub), sehingga memperbarui cabang tersebut dan membagikan pekerjaan dengan tim atau menyinkronkan dengan versi pusat.

